# Forecast-Informed Deep Learning for Electricity Price Prediction in the Australian National Energy Market

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Source code at https://github.com

**Abstract**

Short-term electricity price forecasts support production, purchasing and storage decisions. We compare seven statistical, tree-based and neural forecasting models across the five regions of Australia's National Energy Market, using five-minute data from January 2022 to July 2026. Multivariate models combine engineered market features with P5MIN price forecasts from the Australian Energy Market Operator (AEMO) and eight frozen graph embeddings summarising interregional conditions. A shared pipeline aligns inputs by availability, applies chronological training, validation and test splits, and scores all methods on the same forecast values at twelve leads from 5 to 60 minutes. Evaluation focuses on overall accuracy, with high-price observations retained. All neural models achieve lower mean absolute error (MAE) than persistence and AEMO in every region. The temporal convolutional network (TCN) leads in four regions, while the Transformer leads in Queensland. Regional winners reduce MAE by 15.7–21.4% relative to persistence and 10.7–55.3% relative to AEMO. The multilayer perceptron remains competitive with recurrent models, and TCN fits substantially faster than the Transformer. Rankings vary with the error metric and forecast lead. These findings concern one test period and random seed; matched input ablations are needed to establish the individual contributions of P5MIN forecasts and graph embeddings.

## 1 Introduction

Electricity markets must continually match generation with demand. In Australia's National Energy Market (NEM), the Australian Energy Market Operator (AEMO) dispatches generation every five minutes within network constraints (Australian Energy Market Operator, 2020). Wholesale prices reflect changing supply and demand conditions, making them difficult to anticipate from past prices alone.

Accurate electricity price forecasts help generators plan production, retailers manage purchasing costs and storage operators decide when to charge or discharge. Errors can lead to poorly timed decisions and greater exposure to changing prices. Short-term forecasting therefore matters to participants making operational and trading decisions throughout the day.

This paper applies deep learning to short-term electricity price forecasting. We explore a novel approach that uses AEMO's own P5MIN price forecasts alongside historical market observations as model inputs. These forecasts may provide information about upcoming market conditions that historical prices alone miss. The feature sets reported by Lan, Zhao, and Zhao (2026) and Zhou et al. (2026), discussed in section 2, do not include P5MIN price forecasts.

High-price spikes have historically been important because even brief events can sharply increase wholesale purchasing costs. Retailers often sell electricity at fixed tariffs while buying at variable spot prices, leaving unhedged purchases exposed to large losses. Anticipating spikes therefore helps participants manage purchasing and hedging decisions (Australian Competition and Consumer Commission, 2023).

For Q4 2025, AEMO linked lower volatility and fewer high-price intervals to increased wind and solar output, greater battery discharge and improved coal availability (Australian Energy Market Operator, 2026). High-price intervals are also less frequent in our test period than in 2022 (Figure 1). This makes specialised spike prediction less central to this study, whose focus is accuracy across all intervals. Spikes remain financially important and are retained in training and evaluation.

## 2 Literature Review

Australian electricity price research covers classical machine learning, statistical time-series modelling and deep learning. Sansom, Downs, and Saha (2003) compared support vector machines with neural networks for NSW price forecasting, finding similar accuracy. Apergis et al. (2019) used a hidden semi-Markov model to identify low-price, high-price and spike regimes in historical prices across the five NEM regions.

Recent studies combine several learning architectures. Tan et al. (2023) used signal decomposition with convolutional neural networks and stacked sparse denoising autoencoders for day-ahead NSW price forecasting. Lan, Zhao, and Zhao (2026) combined an inverted Transformer with a bidirectional LSTM, using historical prices, system load, weather and calendar features. Zhou et al. (2026) combined a Kolmogorov–Arnold network with XGBoost for week-ahead forecasting, using lagged prices, demand, net interchange, renewable-generation forecasts and weather. Their different horizons and evaluation settings prevent direct comparisons of reported accuracy.

The input sets reported by Lan, Zhao, and Zhao (2026) and Zhou et al. (2026) do not include AEMO's P5MIN price forecasts. P5MIN provides forecasts at five-minute resolution over the next hour (Australian Energy Market Operator, 2025), incorporating current market bids and offers (Australian Energy Market Operator, 2020). These forecasts could add information about the price-generating process beyond historical observations. We use them as inputs for 5–60 minute predictions and compare the resulting models with AEMO and persistence.

## 3 Dataset and Pre-processing

### 3.1 Dataset

Five-minute market observations and P5MIN forecasts cover the five NEM regions from 1 January 2022 to 31 July 2026. Market observations were collected using NEMOSIS (Gorman et al., 2018) and forecasts using NEMSEER (Prakash, Bruce, and MacGill, 2023). P5MIN forecasts are aligned by publication time, so inputs at origin $t$ contain only information available by $t$.

Table 6 groups the 125 regional inputs and eight graph features shared by the multivariate models. XGBoost also receives the forecast lead. Repeated lags, windows and forecast leads are counted separately. The demand-adjustment field is AEMO's five-minute demand-forecast adjustment (Australian Energy Market Operator, n.d.), rather than a forecast of total demand.

Table 1: Five-minute spot prices by region, January 2022 to July 2026 ($/MWh). The last three columns give the share of intervals below $0, at or above $300 and at or above $1,000.

| State | Median | Mean | Std. dev. | 1st pct. | 99th pct. | $< \$0$ (%) | $\geq \$300$ (%) | $\geq \$1{,}000$ (%) |
|---|---|---|---|---|---|---|---|---|
| NSW | 86.4 | 121.5 | 447 | −45.7 | 475.8 | 7.4 | 4.34 | 0.22 |
| QLD | 85.5 | 115.8 | 422 | −46.8 | 504.0 | 12.6 | 4.67 | 0.29 |
| SA | 71.4 | 103.4 | 494 | −100.5 | 533.9 | 24.7 | 5.35 | 0.29 |
| TAS | 80.4 | 97.2 | 261 | −33.9 | 400.0 | 4.1 | 2.34 | 0.07 |
| VIC | 65.0 | 82.8 | 273 | −65.8 | 449.0 | 20.4 | 3.24 | 0.09 |

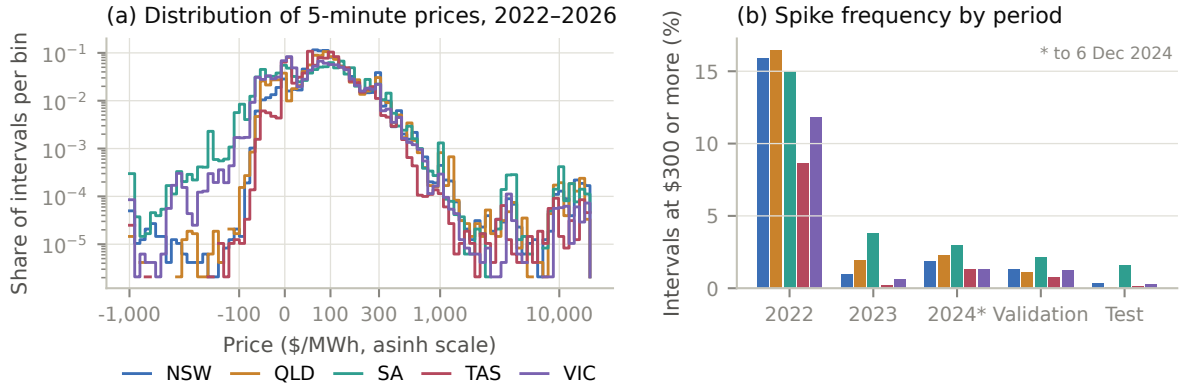



Figure 1: (a) Distribution of 5-minute prices in each region on an asinh scale, with the share of intervals in each bin on a log scale. (b) Share of intervals at or above $300 in each training year, the validation period and the test period. The 2024 bar covers 1 January to 6 December 2024.

### 3.2 Exploratory analysis

Prices are heavy-tailed and can be negative (table 1 and Figure 1); the mean exceeds the median in every region.

High-price events are less common in the test period than in 2022. We retain them and assess overall accuracy.

On VIC training data, current price has a Spearman correlation of 0.97 with its five-minute lag and 0.94 with AEMO's 30-minute price forecast. For the 30-minute price-change target, the correlation is 0.46 with AEMO's forecast change and −0.17 with current price. These relationships motivate forecast inputs and regularisation.

### 3.3 Problem set-up

Let $p_t$ be the price at five-minute interval $t$. Each origin has twelve forecast leads, indexed by $h \in \{1, \ldots, 12\}$, corresponding to 5–60 minutes. Models predict changes and reconstruct prices as

$$y_{t,h} = p_{t+h} - p_t, \qquad \hat{p}_{t+h} = p_t + \hat{y}_{t,h}. \tag{1}$$

Predicting changes makes persistence ($\hat{y}_{t,h} = 0$) a natural reference and reduces dependence on shifts in price level. Sequence models use the last $L$ input vectors, $\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t$; other multivariate models use $\mathbf{x}_t$.

### 3.4 Pre-processing

Neural models fill missing inputs with training-set medians, apply $\operatorname{asinh}(x/100)$ to price inputs and targets, and standardise continuous variables. Asinh handles negative values and heavy tails, unlike an ordinary logarithm or sigmoid function. Calendar encodings are left unscaled. All preprocessing statistics are fitted on eligible training rows and reused for validation and testing. Predictions are inverse-transformed before reconstructing prices. XGBoost uses unscaled inputs and targets; ARIMA applies its own price transformation.

## 4 Experiment Pipeline

A shared pipeline applies the same data splits, forecast leads, selection metric and evaluation rules to every model (Figure 2). New models only need to implement the common training and prediction interface and be registered with their candidate settings. This avoids each team member building a separate workflow and accidentally changing the comparison.

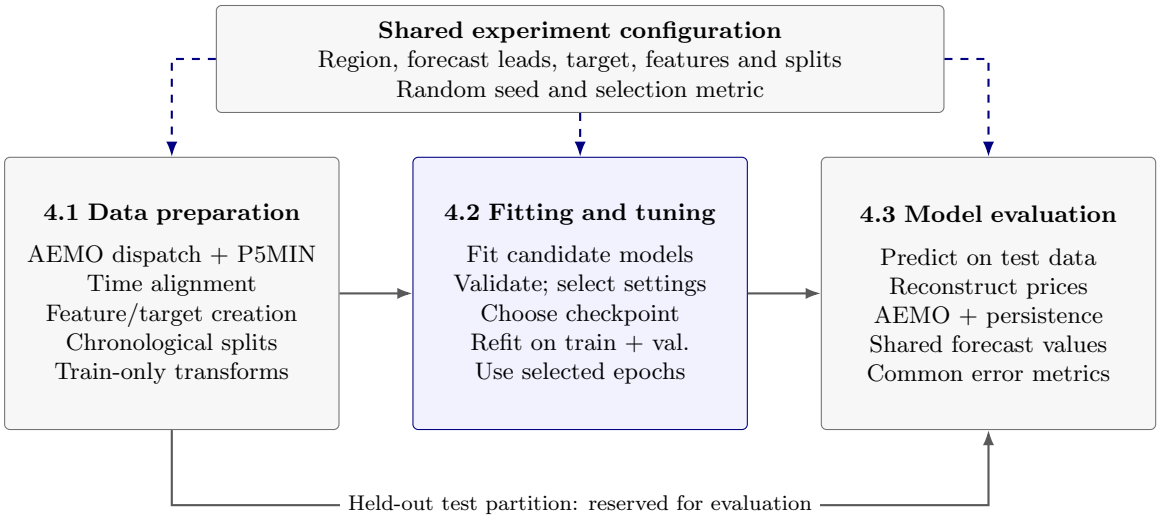



Figure 2: Shared experiment pipeline. Dashed arrows apply common settings; the held-out test partition bypasses model selection.

### 4.1 Data ingestion and preprocessing

The pipeline aligns AEMO observations and forecasts, creates inputs and all twelve targets, and prepares chronological splits. Multivariate models share the engineered inputs, while ARIMA uses regional price history. Sequence windows retain earlier observations across split boundaries, but fitting excludes labels that reach into the next period.

### 4.2 Model fitting and hyperparameter tuning

Each candidate is fitted from scratch and ranked by validation error in the original price units. Neural models share preprocessing, Adam optimisation, batching, seeding and checkpointing. Validation selects the best configuration and neural checkpoint; a fresh final model is then fitted

on training plus validation data, using the selected epoch budget. Architecture, lookback and training settings may differ between models. The test set is excluded from all selection decisions.

### 4.3 Model evaluation

The pipeline reconstructs prices using Equation. (1) and compares them with persistence and AEMO forecasts at matching leads. Every method is scored on the same finite origin–lead pairs. MAE, RMSE, median absolute error and sum of squared errors are computed overall and by lead.

## 5 Experiments

### 5.1 Experimental configuration

We run separate experiments for New South Wales (NSW1), Queensland (QLD1), Victoria (VIC1), South Australia (SA1) and Tasmania (TAS1).Table 5 lists their shared settings; preprocessing follows section 3.4.

Multivariate models share 133 inputs; XGBoost adds the forecast lead to condition its predictions, see (table 6) for a detailed list of features. ARIMA uses regional price history alone. Additional forecast-reliability features are disabled.

**Graph-embedding features.** Graph attention combines information from five region nodes and their interconnector edges. A GRU summarises 12 observations into eight values per region (Figure 3). Training targets are 60-minute price changes and interconnector price differences. The encoder uses the earliest 60% of the pre-test period, with 15% internal validation, and is then frozen. Each forecaster receives its target region's embedding.

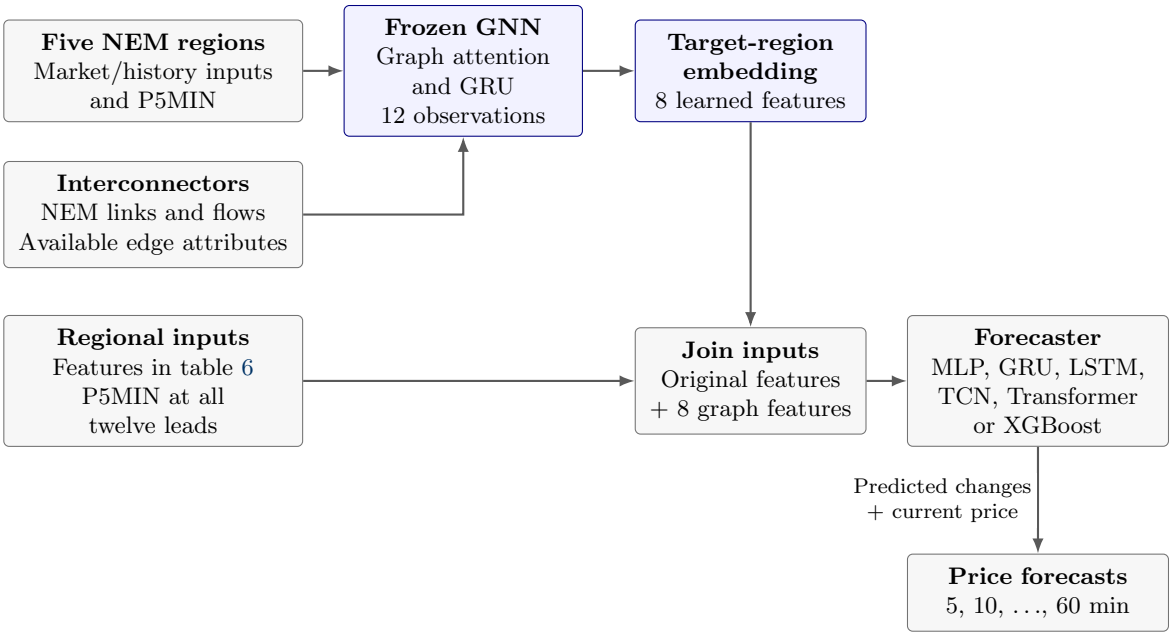



Figure 3: Regional inputs and frozen graph embeddings feed each multivariate model, which predicts twelve price changes for 5–60 minutes ahead.

Each saved experiment evaluates two candidates per model: MAE versus Huber loss for neural models, ARIMA differencing orders 0 versus 1, and XGBoost depths 4 versus 6. Validation MAE selects the candidate and neural refit duration. GRU, LSTM and TCN use 12 observations; the Transformer uses 72. Batch sizes, epoch limits and stopping patience are model-specific.

### 5.2 Experimental results

Table 2 compares all seven models with persistence and AEMO P5MIN forecasts. Within each region, every method is scored on the same 1,155,551 finite forecast values, pooled across test origins and all twelve leads. We report MAE and RMSE in AUD/MWh; RMSE gives greater weight to large errors.

Table 2: Test errors and fit times by region. Errors pool the 5–60 minute forecast leads; bold entries indicate the lowest error for each metric within a region. Fit time includes hyperparameter search and final refitting, excluding shared data preparation and GNN fitting. Dashes indicate unrecorded baseline times.

| | NSW1 | | | QLD1 | | | VIC1 | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | MAE | RMSE | Fit (s) | MAE | RMSE | Fit (s) | MAE | RMSE | Fit (s) |
| TCN | **19.00** | 301.16 | 266.4 | 11.00 | **79.70** | 301.5 | **11.24** | **76.24** | 278.0 |
| Transformer | 19.87 | 293.86 | 3181.8 | **10.56** | 82.31 | 2977.1 | 11.62 | 95.52 | 2519.2 |
| MLP | 19.34 | 301.00 | 266.9 | 10.91 | 87.09 | 282.3 | 11.58 | 95.85 | 188.8 |
| LSTM | 20.59 | 340.97 | 329.2 | 10.95 | 97.61 | 458.1 | 11.47 | 95.35 | 272.4 |
| XGBoost | 19.50 | **289.59** | 894.3 | 11.27 | 81.60 | 914.3 | 12.15 | 90.01 | 939.5 |
| GRU | 21.52 | 343.06 | 266.0 | 11.94 | 103.68 | 408.1 | 11.70 | 96.71 | 163.2 |
| ARIMA | 21.31 | 293.23 | 4.0 | 13.09 | 89.65 | 4.4 | 13.70 | 86.89 | 4.5 |
| Persistence | 23.93 | 353.13 | – | 13.43 | 106.83 | – | 14.10 | 98.83 | – |
| AEMO P5MIN | 42.53 | 638.83 | – | 13.79 | 148.07 | – | 14.26 | 94.25 | – |

| | SA1 | | | TAS1 | | |
|---|---|---|---|---|---|---|
| Model | MAE | RMSE | Fit (s) | MAE | RMSE | Fit (s) |
| TCN | **27.81** | 410.34 | 187.1 | **9.50** | **161.20** | 291.4 |
| Transformer | 28.03 | 389.84 | 4075.9 | 9.95 | 193.67 | 6910.6 |
| MLP | 28.81 | 413.14 | 188.9 | 10.33 | 203.34 | 183.2 |
| LSTM | 29.08 | 421.52 | 384.1 | 10.25 | 207.81 | 337.4 |
| XGBoost | 28.89 | **384.32** | 924.0 | 10.69 | 201.82 | 885.8 |
| GRU | 29.81 | 430.57 | 216.3 | 10.37 | 209.94 | 208.3 |
| ARIMA | 30.92 | 405.35 | 3.9 | 11.01 | 161.23 | 3.7 |
| Persistence | 32.98 | 443.86 | – | 12.07 | 212.10 | – |
| AEMO P5MIN | 31.87 | 409.33 | – | 10.64 | 171.64 | – |

TCN achieves the lowest MAE in four regions, while the Transformer leads in Queensland. These regional winners reduce MAE by 15.7–21.4% relative to persistence and 10.7–55.3% relative to AEMO. Every fitted model improves on persistence in MAE across all five regions. AEMO is a stronger benchmark in Tasmania, where XGBoost and ARIMA have slightly higher MAE than its forecasts.

The RMSE rankings differ: XGBoost performs best in New South Wales and South Australia, while TCN leads in the other three regions. In South Australia, TCN improves on AEMO in MAE but not RMSE, showing that average and large-error performance can differ. TCN also records shorter fit times (187–302 seconds) than the Transformer (2,519–6,911 seconds). Separate experiments removing P5MIN inputs or graph embeddings are needed to identify their individual contributions.

## 5.3 MLP

The multilayer perceptron (MLP) is the simplest network in this study and serves as the reference point for the others. It receives only the input vector at the forecast origin and has no explicit mechanism for modelling sequences; information about the recent past enters only through the engineered lag, change and rolling-window features (table 6). Comparing the MLP with the sequence models therefore shows how much explicit temporal modelling adds beyond the feature engineering.

### 5.3.1 Architecture

The MLP passes the 133 inputs through three fully connected hidden layers of 128, 64 and 32 units. Each hidden layer applies a linear map followed by the GELU activation, and the first two are also normalised and regularised:

$$\mathbf{a}^{(l)} = \mathrm{Dropout}_{p_l}\Big(\mathrm{GELU}\Big(\mathrm{LN}\Big(W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}\Big)\Big)\Big), \qquad \hat{\mathbf{z}}_t = W^{(4)}\mathbf{a}^{(3)} + \mathbf{b}^{(4)} \in \mathbb{R}^{12}, \tag{2}$$

where $\mathbf{a}^{(0)} = \mathbf{x}_t$, $l = 1, 2, 3$ indexes the hidden layers, and layer normalisation (LN) is applied only in the first two. The GELU activation, $\mathrm{GELU}(u) = u\,\Phi(u)$ with $\Phi$ the standard normal cumulative distribution function, is a smooth alternative to the ReLU that passes small negative values rather than cutting them off. Layer normalisation rescales each sample's activations to zero mean and unit variance across the units of a layer, which keeps the scale of the signal stable between layers. Dropout with rates $p_1 = 0.2$ and $p_2 = 0.15$ randomly sets that fraction of units to zero during training, so that the network cannot depend on any single unit. The output layer produces 12 values, the transformed price change at each horizon from 5 to 60 minutes. The network has 28,268 parameters.

### 5.3.2 Hyperparameters

The recorded MLP search compares Huber loss ($\delta = 1$) with MAE on the transformed targets, averaged over the 12 horizons. Validation selects MAE in every region, with final refits of 11–13 epochs. Huber loss is quadratic for small errors and linear for large ones, so it is less sensitive to extreme errors than squared error. Optimisation uses Adam with a learning rate of $10^{-3}$, weight decay of $10^{-4}$ and mini-batches of 256 rows. Training runs for at most 30 epochs and stops once validation MAE has not improved by 0.001 for three epochs, after which the best checkpoint is restored.

## 5.4 TCN

The temporal convolutional network (TCN) models sequences with convolutions rather than recurrence. Unlike the MLP, it receives the last hour of inputs as a sequence, $X_t \in \mathbb{R}^{L \times F}$ with

$L = 12$ five-minute observations and $F = 133$ features per observation, including the frozen graph embeddings in section 5.1. Learned filters slide along this window, so every position is processed in parallel rather than one step after another as in a recurrent network. The receptive field, the span of past observations that can influence an output, is fixed by the architecture rather than learned.

### 5.4.1 Architecture

The building block is a causal dilated convolution. With kernel size $k$ and dilation $d$, the output at window position $s$ is

$$\mathbf{z}_s = \sum_{j=0}^{k-1} W_j \, \mathbf{x}_{s-dj} + \mathbf{b}, \tag{3}$$

where $W_0, \ldots, W_{k-1}$ are learned weight matrices and $\mathbf{b}$ is a bias. The output depends only on the current and earlier positions, which makes the convolution causal. The dilation spaces the filter's weights $d$ positions apart, so it covers a longer span with the same number of weights. Zeros are padded on the past side of the window only, which keeps its length unchanged without letting any output see later observations.

Two residual blocks are stacked, with dilations 1 and 2. Each block applies two causal convolutions with 16 channels and $k = 3$, each followed by a ReLU, and adds the block's input back through a residual (skip/shortcut) connection, $\mathbf{o} = \mathrm{ReLU}\big(g(\mathbf{x}) + W_s\mathbf{x}\big)$. Here $g$ is the pair of convolutions and $W_s$ is a $1 \times 1$ convolution that maps the 133 input features to 16 channels in the first block; the second block uses the identity. The residual path lets each block learn a correction to its input, which makes stacked blocks easier to train. With $B$ blocks the receptive field is

$$r = 1 + 2(k-1)\left(2^B - 1\right) = 13 \text{ observations}, \tag{4}$$

enough to cover the whole 12-observation window. A linear head maps the 16 channels at the latest position to twelve transformed price changes, which are inverse-transformed and added to the current price using Equation. (1). The network has 11,100 parameters, less than half the amount of the MLP (28,268).

### 5.4.2 Hyperparameters

The recorded search fixes the architecture above and compares MAE with Huber loss ($\delta = 1$) on the standardised, asinh-transformed targets, averaged over the twelve leads. Candidate selection and checkpointing use validation MAE after inverse transformation. Training follows section 4.2, using Adam with learning rate $10^{-3}$, weight decay $10^{-4}$, batches of 512, zero dropout and seed 42. The inference batch size is 2,048. Each candidate has a 20-epoch limit and stops after five epochs without a validation MAE improvement exceeding $10^{-3}$. A fresh model is then fitted on training plus validation data for the selected number of epochs. The channel count, kernel size, number of blocks and lookback were not tuned in this search; two blocks are the fewest with $k = 3$ whose receptive field covers the 12-observation window.

## 5.5 Transformer

### 5.5.1 Architecture

Self-attention allows the model to weight historical observations according to the current input, capturing dependencies that may change with market conditions (Wen et al., 2023). In this paper an auto-regressive, decoder only, Transformer was used. This uses sinusoidal positional encoding to create a unique representation for the sequence tokens (Wen et al., 2023).

$$\mathrm{PE}(s, 2i) = \sin(s\, 10000^{-2i/D}), \qquad \mathrm{PE}(s, 2i+1) = \cos(s\, 10000^{-2i/D}), \tag{5}$$

Where $s \in \{0, \ldots, L-1\}$ is the window position, $D = 64$ is the dimensional representation for each token, and $i$ indexes a pair of embedding dimensions. Calendar features separately identify time of day and day of week.
The embeddings then enter a Masked-Multi head attention block. Each head learns query, key and value projections $Q$, $K$ and $V$ of the hidden representation, and computes

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V, \qquad M_{ij} = \begin{cases} 0, & j \le i, \\ -\infty, & j > i. \end{cases} \tag{6}$$

The softmax weights the value vectors by query–key similarity. The causal mask restricts each position to itself and earlier observations; publication-time alignment and chronological splits provide the separate safeguards against data leakage.

Four heads of width $d_k = 16$ are concatenated and projected back to width 64. $N = 3$ decoder blocks layers each apply this attention operation followed by a feedforward network with dimensions $64 \to 128 \to 64$ and a ReLU activation to capture nonlinearities in the higher dimensional projection. Both components use residual connections followed by layer normalization. After a final layer normalization, a linear head maps the last position to twelve transformed price changes. These outputs are predicted jointly, inverse-transformed and added to the current price using Equation. (1).

### 5.5.2 Hyperparameters

The recorded search fixes the architecture above and compares MAE with Huber loss. Both objectives operate on standardised, asinh-transformed targets. Huber loss is quadratic for residuals within $\delta = 1$ and linear beyond that threshold, measured in transformed target units. Candidate selection and checkpointing instead use validation MAE after inverse transformation.

Training follows section 4.2, using Adam with learning rate $10^{-3}$, weight decay $10^{-4}$, batches of 128, zero dropout and seed 42. The inference batch size is 256. Each candidate has a 50-epoch limit and stops after ten epochs without a validation MAE improvement exceeding $10^{-3}$. A fresh model is then fitted on training plus validation data for the selected number of epochs (table 3). The test set does not influence these choices. Architecture and lookback length were not tuned in this search.

Table 3: Transformer loss and refit duration selected by validation MAE.

| Region | NSW1 | QLD1 | VIC1 | SA1 | TAS1 |
|---|---|---|---|---|---|
| Selected loss | Huber ($\delta = 1$) | MAE | MAE | MAE | MAE |
| Refit epochs | 12 | 9 | 7 | 19 | 34 |

### 5.6 Feature analysis using GradientSHAP

We used GradientSHAP to examine which inputs influenced the five neural models' predictions. GradientSHAP estimates how much each input contributes to a model's prediction relative to reference inputs sampled from the fitting data. It does this by measuring the model's sensitivity to each input at random points between the reference and the input being explained. The attribution for input $i$ is

$$\phi_i(\mathbf{x}) = \mathbb{E}_{\mathbf{x}',\alpha}\left[(x_i - x_i')\,\frac{\partial f(\mathbf{z})}{\partial z_i}\bigg|_{\mathbf{z}=\mathbf{x}'+\alpha(\mathbf{x}-\mathbf{x}')}\right], \qquad \alpha \sim U(0,1). \tag{7}$$

Here, $f$ is the inverse-transformed price-change prediction at the forecast lead being explained, $\mathbf{x}$ is the preprocessed model input, $\mathbf{x}'$ is a sampled reference input, and $\alpha$ determines the point along the path. The expectation is estimated using Monte Carlo averaging over sampled references and path positions, with no added input noise. For sequence models, each feature at each time step receives an attribution. These attributions approximately sum to the difference between the predicted price change and the average reference prediction, with positive values indicating an upward contribution and negative values a downward contribution.

The saved explanations contain 128 sampled test origins for each region, model and forecast lead, using reference inputs sampled from the fitting data. We explain the inverse-transformed price-change output, so contributions are measured in AUD/MWh. To summarise feature importance, we take absolute attributions before summing across input positions for sequence models and averaging over origins. This prevents positive and negative contributions from cancelling. We then normalise these magnitudes to sum to 100% within each forecast lead. The resulting shares measure attribution magnitude; they do not retain the direction of each contribution.

Table 4 summarises attribution shares, averaged equally across the twelve leads and five regions. P5MIN price inputs include forecast prices and their differences from the current price; generation and reserve inputs include forecast available generation and reserve margins; demand inputs include forecast demand and its difference from current demand.

Table 4: Mean shares of absolute GradientSHAP attribution (%). Each P5MIN subgroup contains 24 inputs, and the graph group contains eight. The remaining 53 inputs are included in the normalisation but omitted from the table. All P5MIN is the sum of the three P5MIN subgroups, subject to rounding.

| Model | P5MIN price | P5MIN generation and reserves | P5MIN demand | All P5MIN | Graph embeddings |
|---|---|---|---|---|---|
| MLP | 22.4 | 21.2 | 8.5 | 52.1 | 13.3 |
| GRU | 11.1 | 33.6 | 22.8 | 67.5 | 4.9 |
| LSTM | 23.4 | 23.6 | 13.5 | 60.5 | 7.2 |
| TCN | 21.1 | 25.0 | 13.2 | 59.3 | 8.0 |
| Transformer | 22.5 | 21.4 | 10.9 | 54.8 | 7.6 |

P5MIN inputs account for 49.3–70.4% of attribution across the 25 region–model combinations, after averaging over forecast leads. Their role extends beyond the price forecasts themselves. The GRU assigns particularly large shares to forecast generation and reserves (33.6%) and demand (22.8%), compared with 11.1% for forecast price inputs. However, its relatively large P5MIN attribution does not imply better forecasting performance: the GRU has higher MAE than the MLP in every region.

The attribution patterns also vary geographically. In NSW and Queensland, all five neural models assign a larger share to P5MIN generation and reserve inputs than to P5MIN price inputs. In South Australia and Tasmania, four of the five models show the opposite pattern. For TCN, the highest-ranked individual input, averaged across leads, is the 60-minute available-generation forecast in NSW, Queensland and Victoria. In South Australia and Tasmania, it is the five-minute forecast price difference from the current price. These differences indicate that the fitted models use different parts of the forecast information across regions.

Graph embeddings account for 3.6–16.9% of attribution across the region–model combinations, with the largest architecture-level average for the MLP (13.3%). Their share is higher for the 60-minute output than for the five-minute output in all 25 combinations. For TCN, the regional graph shares rise from 3.3–6.4% at five minutes to 8.2–12.8% at 60 minutes. This suggests greater relative use of the learned network summaries at longer forecast leads.

Sequence attribution remains concentrated near the forecast origin. Averaged across leads, the six most recent input rows, spanning 0–25 minutes before the origin, account for 83.2–92.0% of TCN attribution across regions, compared with 65.1–71.7% for the Transformer. These figures describe input-row positions: engineered lags, rolling statistics and graph embeddings can carry older information within a recent row.

These results describe the fitted models' attribution patterns. Absolute shares do not identify the direction of an effect or measure accuracy gains. Correlated inputs and unequal feature-group sizes also affect their interpretation. Mean absolute completeness residuals, averaged across leads, range from 1.2 to 3.2 AUD/MWh across the regional models, indicating that the attribution calculation is approximate. Matched input ablations remain necessary to establish whether P5MIN forecasts and graph embeddings improve predictive accuracy.

### 5.7 Results discussion

The MLP remains competitive with the sequence models despite receiving only the input vector at the forecast origin (section 5.3). Its MAE is lower than the GRU's in every region and lower than the LSTM's in NSW, Queensland and South Australia; in Victoria and Tasmania it is within 1% of the LSTM. With lagged prices, rolling statistics and AEMO forecasts already available as inputs, the MLP provides a strong baseline for assessing explicit sequence modelling. TCN improves on its MAE in four regions, while the MLP narrowly leads in Queensland.

The architecture comparisons also involve a substantial difference in fitting cost. MLP and TCN take roughly 3–5 minutes per region, compared with 42–115 minutes for the Transformer. TCN's short convolutional window therefore offers a useful accuracy–cost trade-off in these experiments. The Transformer's six-hour window gives it access to more history, but this does not consistently improve MAE over the one-hour TCN. These timings include candidate selection and final refitting; differences in epoch budgets, batch sizes and lookback lengths prevent attributing the cost difference to attention alone.

Pooled scores can also hide differences between forecast leads. In Tasmania, AEMO has lower five-minute MAE than the Transformer (3.08 versus 4.66 AUD/MWh), while the Transformer performs better at 60 minutes (12.26 versus 16.03 AUD/MWh). Its lower overall MAE therefore does not imply an improvement at every lead. Together with the different MAE and RMSE rankings in table 2, this suggests that model choice should depend on the required lead and the importance placed on large errors.

These comparisons use one random seed and one chronological test period, without uncertainty

intervals, so small differences should be interpreted cautiously. They evaluate models using their complete input sets and do not establish how much of the improvement comes from P5MIN forecasts or graph embeddings. Experiments removing each input group, repeated training seeds and additional test periods would help separate these effects and assess whether the rankings persist under different market conditions.

# 6 Conclusion

We compared seven electricity price forecasting models across the five NEM regions at leads from 5 to 60 minutes. The multivariate models combined historical market observations with AEMO's P5MIN forecasts and frozen graph embeddings. A shared pipeline applied consistent chronological splits, preprocessing rules and evaluation criteria, allowing the models to be compared on the same forecast values.

All neural models achieved lower MAE than both persistence and AEMO in every region. TCN led in four regions, while the Transformer led in Queensland. The regional winners reduced MAE by 15.7–21.4% relative to persistence and 10.7–55.3% relative to AEMO (table 2). The MLP remained competitive with recurrent models, and TCN offered a favourable balance between accuracy and fitting cost. However, rankings changed with the error metric and forecast lead, so model choice should reflect the intended forecasting task.

## 6.1 Limitations

Evaluation used one chronological test period and one random seed. High-price intervals were less frequent in the test period than in 2022, so the results do not establish robustness under more volatile conditions. Overlapping forecast origins and leads also produce dependent errors; the number of scored values should not be interpreted as a count of independent replications. Repeated seeds, further test periods and uncertainty estimates that account for temporal dependence would give a clearer picture of ranking stability.

The recorded search compared only two candidates per model, with most architecture settings fixed. Lookback lengths, batch sizes and epoch budgets also differed, limiting conclusions about architectural superiority and the causes of fitting-cost differences. Matched experiments removing P5MIN forecasts and graph embeddings are needed to identify their individual contributions. The observed gains belong to the complete fitted models and cannot be attributed to either input group alone.

Finally, the reported metrics assess point-forecast accuracy. Live forecasting performance, predictive uncertainty and the value of forecasts for trading or storage decisions require separate evaluation.

# Appendix

Table 5: Shared experiment settings. Split proportions are approximate, before excluding labels that cross boundaries.

| Setting | Value |
|---|---|
| Training/validation/test | Chronological 64%/16%/20% split. |
| Validation | One chronological holdout; a 60-minute label embargo separates fitting from validation. |
| Selection metric | MAE pooled across forecast origins and all twelve leads. |
| Random seed | 42. |
| Neural optimiser | Adam; learning rate $10^{-3}$ and weight decay $10^{-4}$. |
| Benchmarks | Persistence and AEMO P5MIN prices at the corresponding forecast leads. |

Table 6: Inputs before scaling; counts are in parentheses. Times are in minutes. Each P5MIN input has twelve variants, at 5, 10, . . . , 60 minutes ahead.

| Feature name | Scale | Description |
|---|---|---|
| Regional reference price | \$/MWh | Current regional spot price. |
| Total demand | MW | Current regional demand. |
| Available generation | MW | Available generating capacity. |
| Demand adjustment | MW | Five-minute demand-forecast adjustment. |
| Dispatchable generation | MW | Dispatched generation. |
| Dispatchable load | MW | Dispatched load. |
| Net interchange | MW | Net transfer between regions. |
| Reserve margin | MW | Available generation minus demand. |
| Reserve ratio | Unitless | Reserve margin divided by demand. |
| Demand difference | MW | Demand minus the demand adjustment. |
| Lagged price (7) | \$/MWh | Price 5, 10, 15, 30, 60, 180, 1,440 minutes earlier. |
| Lagged demand (5) | MW | Demand 5, 15, 30, 60, 1,440 minutes earlier. |
| Price change (4) | \$/MWh | Change over 5, 15, 30, 60 minutes. |
| Demand change (4) | MW | Change over 5, 15, 30, 60 minutes. |
| Rolling price mean (4) | \$/MWh | Mean over 30, 60, 180, 1,440 minutes. |
| Rolling price std. dev. (4) | \$/MWh | Standard deviation over the same windows. |
| Rolling demand mean (4) | MW | Mean demand over the same windows. |
| Interconnector flow (6) | MW | Signed flows on six NEM interconnectors. |
| P5MIN price (12) | \$/MWh | Forecast regional price. |
| P5MIN demand (12) | MW | Forecast total demand. |
| P5MIN generation (12) | MW | Forecast available generation. |
| P5MIN price change (12) | \$/MWh | Forecast price minus current price. |
| P5MIN demand change (12) | MW | Forecast demand minus current demand. |
| P5MIN reserve margin (12) | MW | Forecast generation minus forecast demand. |
| Time of day (2) | $[-1, 1]$ | Sine and cosine of time within the day. |
| Day of week (2) | $[-1, 1]$ | Sine and cosine of day within the week. |
| Weekend indicator | $\{0, 1\}$ | One on weekends; zero otherwise. |
| Graph embedding (8) | Learned | Target-region summary of the NEM network. |
| Forecast lead (XGBoost) | Minutes | Lead being predicted, from 5 to 60 minutes. |

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